

Methodology

Phase 1: Site identification and paired design

Documentary review

Satellite photointerpretation

Active brick kiln polygon

Paired control selection

Pair distance validation

Phase 2: Remote sensing analysis

Landsat 8 OLI/TIRS

Cloud masking

Temporal compositing

8 spectral / thermal indices computed per image

LST

NDVI

NDMI

BSI

UI

MNDWI

IBI

NDBI

Multi-buffer pixel extraction

Phase 3: Statistical analysis

Shapiro-Wilk

Wilcoxon signed-rank

Spearman ρ + gradient